



Introduction to Algebra — Course Syllabus

Mary, Star of the Sea School · SY 2026-2027

Course Description

Math 8 is a full high-school Algebra 1 course. You will work with linear functions, systems of equations, exponents and scientific notation, the Pythagorean theorem, and exponential and inverse-variation relationships — all the algebra needed for a confident start in high-school math. The course is paced to support STEM class throughout the year: chemistry uses linear equations and weighted averages, motion labs use physics equations of motion and trigonometry, and the spring electricity-and-waves unit relies on Ohm's Law, inverse variation, and Kepler's geometry. Class time balances skill-building practice, project-based summatives, lab work, and direct connection to real Hawaiian and engineering problems.

Textbook: Glencoe Math Algebra 1

Teacher

Dr. Peter Park

✉ ppark@starofthesea.org

☎ (808) 228-2285

Best method: Email is fastest. I respond within 24 hours on school days.

To schedule a meeting: Email me with 2-3 times that work — I can meet before

school (7:30–7:50 a.m.), during my prep period, or after school by appointment.

Phone: Leave a message with the school office and I will return the call within one school day.

Materials

Provided by the school: textbook (Glencoe Math Algebra 1), classroom manipulatives, and in-class technology devices.

Students bring daily:

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 - Composition notebook
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 - Pencils, erasers, highlighters
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 - Notebook paper
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 - Basic 4-function or scientific calculator
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 - Folder for handouts
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Later in the year: graph paper, ruler/protractor, and a small compass are needed for the geometry unit — a supply reminder will go home in advance.

Grading Policy

20%ParticipationComposition books, notes, in-class behavior, supplies

25%TestsQ&A, written essay (1 per chapter, 20 pts)

30%HomeworkAssignments, classwork, class prep, projects

25%Quizzes2 per chapter, 5 pts each

Make-ups: Students who score below 80% on tests may make corrections / retest to improve their grade.

Late work: Homework is due the next class meeting. Late work earns a maximum of 50% credit; if not turned in after two class meetings, the grade is a zero.

Standardized testing: Students take the STAR assessment in fall, winter, and spring to track math growth. English-language learners take the WIDA assessment as required.

Parent-teacher conferences: Formal conferences are held after Quarter 1 and Quarter 3 grades post. Additional conferences may be requested by email anytime.

Course Outline by Quarter — Introduction to Algebra

□ View Math ↔ Physical Science Alignment Chart		□ Glencoe
Textbook Pacing Guide	□ Summative Projects Guide	□ Math ↔
STEM Co-Assessment Map	□ Cross-Curricular Projects	

| Q1 — Lab Math & Number Foundation

- | •Lab Math Refresher: Significant Figures, Percent Error, Scientific Notation w/ computation, Unit Conversions, Density review
- | •Real Numbers; Square Roots
- | •Exponents & Laws of Exponents

| Q2 — Lines, Slopes & Stoichiometry

- | •Slope & Rate of Change
- | • **Linear Equations** (Standard, Slope-Intercept)
- | •Rearranging Formulas (essential for stoichiometry)
- | •Systems of Equations — ready for STEM chemistry unit
- | •**Percent & Weighted Averages** — for the STEM Element Babybook (isotope abundance) and Stain Lab (concentrations)

Q3 — Math for Motion (Front-Loaded)

- **Physics Equations of Motion** — taught early January for STEM Velocity / Acceleration / Projectile Labs
- Functions; Linear Functions
- Pythagorean Theorem; Intro to Trigonometry (SOH-CAH-TOA)
- Transformations; Angle Relationships

Q4 — Electricity, Waves & Cosmos

- Ohm's Law & Linear Functions; Series and Parallel Circuits
- Waves — Frequency & Wavelength (inverse variation)
- Optics & Snell's Law (uses Q3 trig)
- Kepler's Laws of Orbital Mechanics
- Volume; Scatter Plots; Two-Way Tables; Word Problems capstone

Student & Parent Communication

Beehively is the platform where students and parents can regularly check grade progress and receive notifications of missing or incomplete assignments.

Parents are contacted by email up to 2 times if their child is receiving a deficiency, plus standard email notifications. Students are always informed of any grade that is a C or lower and ultimately are responsible for their grades.

Student work is updated regularly.

Middle School Math Goal

From 6th through 8th grade, we focus on more than just numbers — we work to build strong math skills, boost confidence, and develop critical problem-solving abilities that students can apply in every subject and in everyday life. Our approach combines hands-on activities, real-world applications, and a supportive learning environment where students feel encouraged to take risks, ask questions, and think

creatively.

By the time they graduate, our students are not only ready for Algebra and beyond, but they also possess the perseverance, analytical thinking, and resourcefulness they need to thrive in high school and in life.

Schoolwide Learning Expectations (SLEs): Self-Aware, Confident Individuals; Teachable Life-Long Learners; Active Christians; Responsible Citizens.

Daily Schedule

Please refer to the MSOS Middle School daily bell schedule (a hard copy is included in your first-week folder) for exact times. Highlights:

- **School day:** Morning arrival, morning prayer, and classes begin per the master schedule.
 - **Recess & lunch:** Mid-morning recess and midday lunch as posted.
 - **Specials:** P.E., Music, Art, Library, and Chapel rotate on specific days — please check the weekly schedule sent home.
 - **Dismissal:** End-of-day dismissal per the master schedule; pickup procedures are in the family handbook.
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Homework & Study Expectations

Frequency: Math homework is assigned most class meetings. Middle-school

students should expect roughly **20-30 minutes of math per night** (part of the ~60-80 minutes total nightly homework recommended for grades 6-8).

Ongoing practice: Composition notebooks are updated every class. Students should also spend a few minutes on math fluency (IXL, Khan Academy, or flash cards) between formal assignments.

Tips for supporting math at home:

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 - Set a consistent, quiet study time and place.
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 - Ask your child to explain the problem back to you — teaching cements learning.
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 - Encourage them to attempt every problem, even imperfectly; productive struggle is part of learning math.
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 - Contact me early if a topic is causing frustration — small course-corrections are much easier than late-quarter recoveries.

Late-work policy is described in the Grading Policy above.

Classroom Procedures & Behavior Expectations

Our three class rules — the 3 R's:

- **Respect** — ourselves, one another, our teachers, and our learning space.
- **Responsibility** — arrive on time, prepared with materials, and ready to work.

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- **Readiness** — participate fully, ask questions, and give your honest best effort.

Positive recognition: Students who consistently show the 3 R's earn verbal praise, "shout-outs" in class, positive Beehively notes home, and class-wide rewards (extra math games, free-choice problems, review-day celebrations).

When expectations aren't met: I always start with a quiet private conversation. If a behavior continues, the sequence is (1) private check-in and reset, (2) reflection or written response during class, (3) parent contact, and (4) a meeting with the family and principal if needed. The goal is always to redirect and re-teach — not to punish.

Technology Use

Platforms used in this class:

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- **Beehively** — grade portal and family notifications.

 - **Google Classroom** — assignment posting, digital worksheets, and turn-in.

 - **IXL** — targeted skill practice for every unit.

 - **Khan Academy / YouTube** — supplemental video instruction linked in each unit.

 - **Class iPads / Chromebooks** — used in class for interactive activities and research.

Digital citizenship: All device use follows the MSOS *Technology Acceptable*

Usage Policy in the family handbook. Students are expected to use school devices only for schoolwork, respect classmates online, protect their passwords, and never share personal information. Any misuse is handled per the school's AUP.

Room Parent & Parent Volunteers

Ways to help: chaperoning field trips, guest-speaking about math in your profession, helping with class events (Pi Day, Math Olympics, STEM Fair), and providing occasional classroom supplies.

How to sign up: Please email me if you're interested and I'll connect you with our room parent. All volunteers must have current safe-environment training on file with the school office.

Key dates to remember: Back-to-School Night, Parent-Teacher Conferences (after Q1 and Q3), STEM Fair, Pi Day (March 14), and end-of-year Math Awards. Exact dates are published on the school calendar and in weekly newsletters.

How to Support Your Child at Home

- **Routines:** Build a consistent homework time and a distraction-free spot. Middle-schoolers do best with a predictable rhythm.
- **Independence:** Let your child own the process — check that homework is attempted, but let mistakes happen so we can learn from them in class the next day.

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- **Communication:** Check Beehively weekly with your student. Celebrate wins and unpack misses together.
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- **Real-world math:** Point out math in cooking, sports stats, shopping, driving, and budgeting. Math is everywhere on Kaua'i.
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- **Growth mindset:** Praise effort and strategy (“you kept trying different approaches”) more than results.
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Final Notes & Acknowledgment

Thank you for partnering with me this year. Your support at home is the single biggest factor in your child's confidence and success in math. I am grateful for the trust you place in Mary, Star of the Sea and in this class.

Class website: starofthesea.org/classes — unit pages, worksheets, quizzes, and this syllabus are all live there.

Parent Acknowledgment: After reviewing this syllabus with your student, please email ppark@starofthesea.org with the subject line “Math 8 Syllabus Reviewed — [Student Name]” so I know it has reached your family. Mahalo!